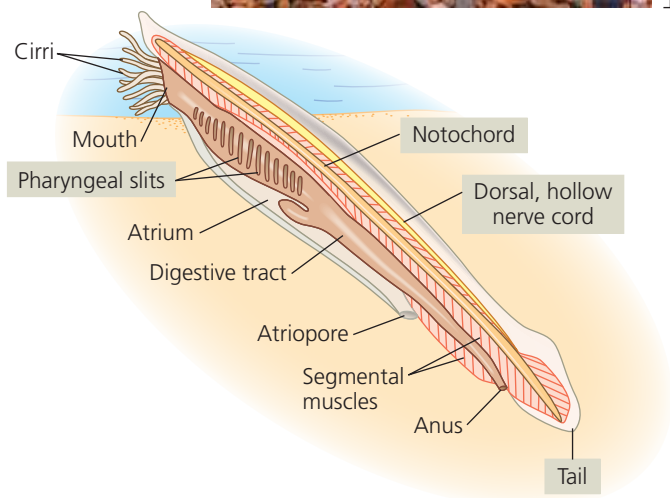
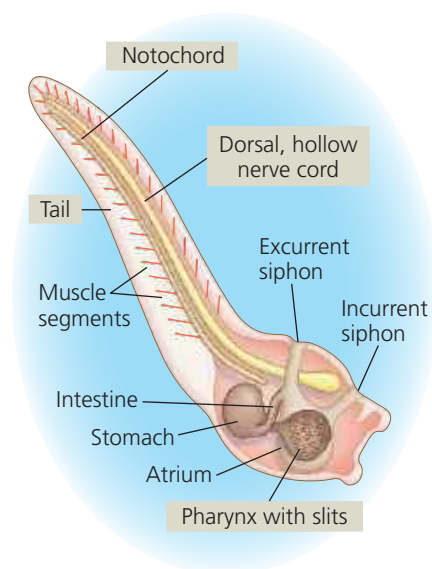


▼ **Figure 34.4** The lancelet *Branchiostoma*, a cephalochordate.

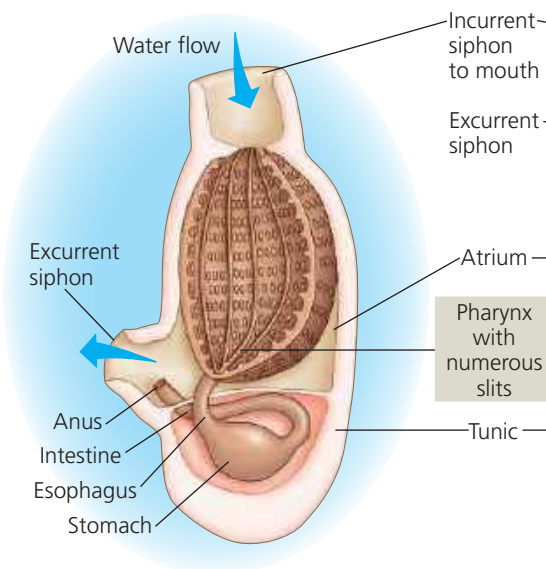
This small invertebrate displays all four main chordate characters. Water enters the mouth and passes through the pharyngeal slits into the atrium, a chamber that vents to the outside via the atriopore; large particles are blocked from entering the mouth by tentacle-like cirri. The serially arranged segmental muscles produce the lancelet's wavelike swimming movements.



▼ **Figure 34.5** A tunicate, a urochordate.



(a) A tunicate larva is a free-swimming but nonfeeding "tadpole" in which all four main characters of chordates are evident.



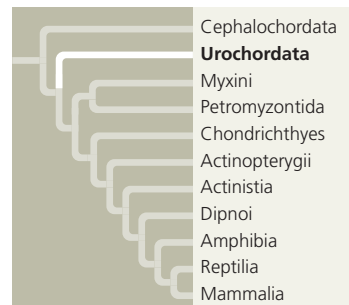
(b) In the adult, prominent pharyngeal slits function in suspension feeding, but other chordate characters are not obvious.



(c) An adult tunicate, or sea squirt, is a sessile animal (photo is approximately life-sized).

Globally, lancelets are rare, but in a few areas (such as Tampa Bay, on the Florida coast), they may reach densities of more than 5,000 individuals per square meter.

Tunicates



Recent molecular studies indicate that the **tunicates** (Urochordata) are more closely related to other chordates than are lancelets. The chordate characters of tunicates are most apparent during their larval stage, which may be as brief as a few minutes

(Figure 34.5a). In many species, the larva uses its tail muscles and notochord to swim through water in search of a suitable substrate on which it can settle, guided by cues it receives from light- and gravity-sensitive cells.

Once a tunicate has settled on a substrate, it undergoes a radical metamorphosis in which many of its chordate characters disappear. Its tail and notochord are resorbed; its nervous system degenerates; and its remaining organs rotate 90°. As an adult, a tunicate draws in water through an incurrent siphon; the water then passes through the pharyngeal slits into a chamber called the atrium and exits through an excurrent siphon (Figure 34.5b and c). Food particles are filtered from the water by a mucous net and transported by cilia to the esophagus. The anus empties into the excurrent siphon. Some tunicate species shoot a jet of water through their excurrent siphon when attacked, earning them the informal name of "sea squirts."